

Lesson 12 · Signed demand elasticity and revenue

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If an airline increases prices for business class flights by 8%, demand falls by about 2.5%. Estimate the signed elasticity of demand.

☐ A 0.3125☐ B -3.2☐ C -0.3125☐ D -2.5

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Q2 · Rank 1 / Foundation

Find the signed price elasticity of demand at $P = 6$ for $P = 30 - 2Q$.

☐ A -0.25☐ B -4☐ C 0.25☐ D -0.5

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Q3 · Rank 1 / Foundation

Find the signed price elasticity of demand at $P = 6$ for $P = 30 - 12Q$.

☐ A $-1/12$

☐ B -1.5

☐ C 0.25

☐ D -0.25

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Q4 · Rank 2 / Core

If the demand equation is $Q + 4P = 60$, find a general expression for the signed price elasticity in terms of P , for $0 < P < 15$.

☐ A -4

☐ B $-P/(15 - P)$

☐ C $P/(15 - P)$

☐ D $-(15 - P)/P$

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Q5 · Rank 2 / Core

If the demand equation is $Q + 4P = 60$, for what value of P is demand unit elastic?

☐ A 15☐ B 30☐ C 7.5☐ D 0

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Q6 · Rank 2 / Core

Find the signed elasticity of demand in terms of Q for $P = 20 - 0.05Q$, for $0 < Q < 400$.

☐ A $-(400 - Q)/Q$ ☐ B -0.05 ☐ C $-Q/(400 - Q)$ ☐ D $+(400 - Q)/Q$

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Q7 · Rank 3 / Application

For demand $P = 20 - 0.05Q$, at what value of Q is demand unit elastic?

☐ A 400☐ B 20☐ C 10☐ D 200

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Q8 · Rank 3 / Application

For demand $P = 20 - 0.05Q$, find marginal revenue and evaluate it where demand is unit elastic. Choose (MR , value at that point).

☐ A $(20 - 0.05Q, 10)$ ☐ B $(20 - 0.1Q, 0)$ ☐ C $(20Q - 0.05Q^2, 2000)$ ☐ D $(20 - 0.1Q, 20)$

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Q9 · Rank 3 / Application

An airline raises business class prices by 8% and demand falls by about 2.5%. Is demand elastic, inelastic or unit elastic?

- | | |
|--|--|
| ☐ Elastic, because price rose more than demand fell.
A | ☐ Unit elastic, because both quantities changed.
B |
| ☐ C Inelastic, because $ E = 0.3125 < 1$. | ☐ D Undefined, because elasticity is negative. |

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Q10 · Rank 4 / Challenge · OPTIONAL

Demand is $Q = 1000/P^2$. Estimate the percentage change in demand when price increases by 2% from 5, then compare it with the exact change when price rises from 5 to 5.1. Choose (estimate, exact), rounded to two decimal places.

- | | |
|--|--|
| ☐ A (−4%, −3.88%) | ☐ B (−2%, −3.88%) |
| ☐ C (−4%, −4%) | ☐ D (4%, 3.88%) |

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